

OpenStax Chemistry 2e Errata

<https://openstax.org/details/books/chemistry-2e>

May 12, 2026 – Dean Peterson

Notes:

- I am a chemistry learner, not an expert, doing this avocationally not in a class setting.
- Since I lack instructor access, I could not review even numbered exercise solutions.
- I have attempted to include screenshots of unlabeled figures to assist in locating them.
- This document should have a PDF table of contents created from the headings.

Chapter 8

Possible Incomplete Figure Updates

This issue happened a lot in Chapter 8 but it is not limited to chapter 8. I noticed a lot of text referring to what I assume are old versions of figures. The issue cropped up in figure alt text, figure captions, and even in normal non-caption text in a couple of places. I could be wrong that these are references to old versions of figures, but to give an example, there are references to purple orbitals in multiple alt texts even though none of the figures include purple orbitals.

I mention this in case OpenStax wants to check on procedures for getting alt text, captions, and main text updated when a figure is updated. Here are non-exhaustive lists of examples that demonstrate what I am talking about. Specific errata for these issues appear later in this document.

In normal text:

- Chapter 8.2, section sp² Hybridization, 1st paragraph after figure 8.10
- Example 8.2, Solution section, 1st sentence wording

In figure captions:

- Figure 8.2
- Figure 8.8
- Figure 8.10
- Figure 8.15

In alt text:

- Figure 8.2
- Figure 8.7
- Figure 8.15
- Unlabeled figure just above Example 8.2

- Figure 8.23
- Figure 8.24
- Figure 8.25

Chapter 8 - Introduction

Figure 8.1 alt text

The alt text reads (emphases mine):

A pitcher is shown pouring liquid oxygen through the gap between two magnets, where it **has formed a solid disk**. A call out box near the stream of liquid oxygen shows an image of **six** pairs of spheres, **spread apart from one another**. Another call out box near the solid disk shows **ten** pairs of spheres **much closer together**.

Three issues, the first of which significantly affects the meaning of the figure:

- The distinction between the call out boxes is not, as stated, how close together the oxygen atoms are. The distinction is whether or not the oxygen atoms are all aligned the same way in the magnetic field.
- The phrase “has formed a solid disk” gives the impression that the liquid oxygen becomes solid between the magnets.
- The counts of spheres in the call out boxes are wrong.

Here is suggested replacement alt text (changes in bold):

A pitcher is shown pouring liquid oxygen through the gap between two magnets, where it **pools even though there is nothing below it to hold it up**. A call out box near the stream of liquid oxygen shows an image of **seven** pairs of spheres, **all oriented in random directions**. Another call out box near the solid disk shows **seven** pairs of spheres **all oriented in the same direction with one sphere of the pair to the left and the other to the right**.

Chapter 8.1

Figure 8.2 caption

The caption includes part letters “(a)” and “(b)” even though the figure does not. Suggest removing the part letters from the caption.

Figure 8.2 alt text

The alt text reads:

A pair of diagrams are shown and labeled “a” and “b”. Diagram a shows three consecutive images. The first image depicts two separated blurry circles, each labeled with a positive sign and the term “H atom.” The phrase written under them reads,

“Sufficiently far apart to have no interaction.” The second image shows the same two circles, but this time they are much closer together and are labeled, “Atoms begin to interact as they move closer together.” The third image shows the two circles overlapping, labeled, “H subscript 2,” and, “Optimum distance to achieve lowest overall energy of system.” Diagram b shows a graph on which the y-axis is labeled “Energy (J),” and the x-axis is labeled, “Internuclear distance (p m).” The midpoint of the y-axis is labeled as zero. The curve on the graph begins at zero p m and high on the y-axis. The graph slopes downward steeply to a point far below the zero joule line on the y-axis and the lowest point reads “0.74 p m” and “H bonded to H bond length.” It is also labeled “negative 7.24 times 10 superscript negative 19 J.” The graph then rises again to zero J. The graph is accompanied by the same images from diagram a; the first image correlates to the point in the graph where it crosses the zero point on the y-axis, the third image where the graph is lowest.

Issues:

- The alt text must describe an old figure where the fuzzy hydrogens and the graph were separate images, parts (a) and (b). The current figure only has the graph, and the fuzzy hydrogens are shown above the curve of the graph on the same image.
- The H-H bond length is given as 0.74 pm. In fact, it is 74 pm (it would be 0.74 in Angstroms)
- Minor things like mentioning positive signs that are not on the current version of the figure, labels being above versus below, etc.

Suggested replacement alt text:

The diagram shows a graph on which the y-axis is labeled “Energy (J),” and the x-axis is labeled, “Internuclear distance (p m).” The midpoint of the y-axis is labeled as zero. The curve on the graph begins at zero p m on the x-axis and high above zero on the y-axis. The graph slopes downward steeply to a point far below the zero joule line on the y-axis. The lowest point reads “74 p m” and “H bonded to H bond length” on the x-axis. It is also labeled “negative 7.24 times 10 superscript negative 19 J” on the y-axis. The graph then rises again more slowly, flattening out as it rises to approach zero J. Above the graph are a series of three images. On the far right, where the curve is flat and near zero J, an image depicts two separated blurry circles, each labeled with the symbol “H.” A phrase written above them reads, “Sufficiently far apart to have no interaction.” Farther left, where the graph is getting lower but still to the right of its lowest point, an image shows the same two circles, but this time they are much closer together and are labeled, “Atoms begin to interact as they move closer together.” Even farther to the left, above the graph’s lowest point, an image shows the two blurry circles overlapping, labeled, “H subscript 2,” and, “Optimum distance to achieve lowest overall energy of system.”

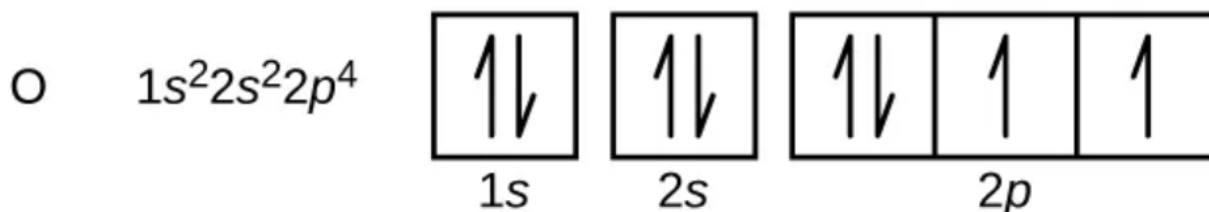
Chapter 8.2

Chapter 8.2, section Learning Objectives, second unbulleted paragraph

Sentence 4 reads (emphasis mine):

Oxygen has the electron configuration $1s^2 2s^2 2p^4$, with two unpaired electrons (one in each of the **two** $2p$ orbitals).

This makes it sound like oxygen has two p orbitals, not three. I checked the orbital diagrams in chapter 6.4, section Aufbau Principle, and found the diagram for oxygen (screenshot below.) It contains three p orbitals:



Maybe sentence 4 would be better worded as (change in bold):

Oxygen has the electron configuration $1s^2 2s^2 2p^4$, with two unpaired electrons (**one each in two of the three $2p$ orbitals**).

Another rephrasing might be (change in bold):

Oxygen has the electron configuration $1s^2 2s^2 2p^4$, with two unpaired electrons (one in each of the two **incompletely filled $2p$ orbitals**).

Figure 8.7 alt text

The alt text reads:

Two diagrams are shown and labeled “a” and “b.” Diagram a shows two peanut-shaped orbitals lying in a tetrahedral arrangement around the letter “O.” Diagram b shows the same two orbitals, but they now overlap to the top and to the left with two spherical orbitals, each labeled “H.” A pair of electrons occupies each lobe of the peanut-shaped orbitals.

This description appears old. A couple of issues.

- There is no mention of the different angles labeled on parts (a) and (b).
- Is “peanut-shaped” really the best shape description for orbitals arranged in a tetrahedron? They would be very bent peanuts.
- There are no letters “O” or “H”.
- The spherical orbitals are described as being to the top and left (like figure 8.6), not bottom and right like they are in this figure.

In short, this alt text is probably old and should be revised.

Chapter 8.2, section Learning Objectives, numbered point 6

At the end of the Learning Objectives section are numbered points about hybrid orbitals. Point 6 reads:

6. Hybrid orbitals overlap to form σ bonds. Unhybridized orbitals overlap to form π bonds.

Is the second sentence strictly true? Chapter 8.1 describes both σ and π bonds that form from unhybridized orbitals. However, I may be misunderstanding some context. Would it be more accurate to say this?

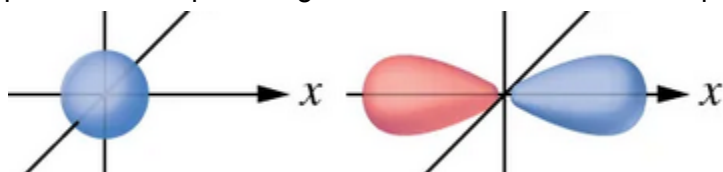
6. Hybrid orbitals **can** overlap to form σ bonds. **On the other hand, π bonds can only form from unhybridized orbitals.**

Figure 8.8 caption

The first sentence of the figure 8.8 caption reads:

Hybridization of an s orbital (blue) and a p orbital (red) of the same atom produces two sp hybrid orbitals (yellow).

Can someone check that first sentence for accuracy? My interpretation of that part of the picture is that the blue sphere is an s orbital, and that the red and blue lobes together make up a single p orbital. The part of figure 8.8 I'm worried about is reproduced below.



The figure 8.10 and 8.15 captions have similar issues.

Figure 8.8 alt text

The alt text looks out of date. Issues include discussion of a “red, peanut-shaped orbital” when the peanut shape is partly red and partly blue, and a mention of a purple orbital when there is no purple orbital on the diagram.

Figure 8.9 alt text

The alt text reads:

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left diagram shows an up-facing arrow labeled, “E.” To the lower right of the arrow is a short, horizontal line labeled, “2 s,” that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, “2 p.” Above these two sets of lines is the phrase, “Orbitals in an isolated B e atom.” The right side of the diagram shows two short, horizontal lines placed halfway up the space and each labeled, “s p.” An upward-facing half arrow is drawn vertically on each line. Above these lines are two other short, horizontal lines, each labeled, “2 p.” Above these two sets of lines is the phrase, “Orbitals in the s p hybridized B e in B e C l subscript 2.”

This alt text is correct, but I think it could be clearer. Things it omits or only implies include:

- That the “s p” orbitals on the right side are at the same height/energy level.
- That the “2 p” orbitals on the right side are the same height as each other.

- That the “2 p” orbitals on the left and right sides are at the same height.
- That two orbitals were replaced and two remained unchanged.
- The sides are referred to inconsistently as “The left diagram” and “The right side of the diagram” (minor.)

Here is a possible replacement (changes in bold):

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left **side of the** diagram shows an up-facing arrow labeled, “E.” To the lower right of the arrow is a short, horizontal line labeled, “2 s,” that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, “2 p.” **The left side of the diagram is labeled,** “Orbitals in an isolated B e atom.” The right side of the diagram **still has two of the three “2 p” lines at the top, but it replaces the third “2 p” line and the low “2 s” line with two new lines on the same level midway up the diagram. The two new lines are labeled,** “s p.” An upward-facing half arrow is drawn vertically on each line. **The right side of the diagram is labeled,** “Orbitals in the s p hybridized B e in B e C l subscript 2.”

Figure 8.10 caption

The first sentence of the figure 8.10 caption starts:

The hybridization of an s orbital (blue) and two p orbitals (red) ...

The p orbitals are not entirely red. I believe this is the same issue as the figure 8.8 caption.

Figure 8.10 alt text

The alt text reads:

This shows a series of three diagrams with one on the left connected to one on the right by a right-facing arrow that is labeled, “Hybridization.” Below the one on the right is a downward-facing arrow labeled, “Gives a trigonal planar arrangement,” connecting to the last diagram. The first diagram shows a blue spherical orbital labeled “S” and then two red and blue, peanut-shaped orbitals, each placed on an X, Y, Z axis system, labeled “P subscript x” and “P subscript y.” The two red and blue orbitals are located on the x and z axes, respectively. The second diagram shows the three orbitals again on an X, Y, Z axis system, but they are yellow and have one enlarged lobe and one smaller lobe. Each lies in a different axis in the drawing. The third diagram shows the same three orbitals, but their smaller lobes now overlap while their larger lobes are located at and labeled as “120 degrees” from one another.

Issues:

- Sentence 4 says the p orbitals are along the x and z axes. They are along the x and y axes, consistent with the “P subscript y” in the previous sentence.
- Sentence 5 says the second diagram “shows the three orbitals again”, implying that it is showing the s and two p orbitals again. It is not.
- Sentence 6 says that in the second diagram, the hybrid orbitals “Each lies in a different axis in the drawing.” This is not the clearest way to say they are separate images on

separate coordinate systems. Moreover, it gives the impression that the three hybrid orbitals lie along the x, y, and z axes, which they do not.

- Sentence 7 says of the third diagram that “their smaller lobes now overlap” giving the impression that the smaller lobes overlap with each other. They do not. They overlap with the larger lobes.

Here is a suggested revision (changes in bold):

This shows a series of three diagrams with one on the left connected to one on the right by a right-facing arrow that is labeled, “Hybridization.” Below the one on the right is a downward-facing arrow labeled, “Gives a trigonal planar arrangement,” connecting to the last diagram. The first diagram shows a blue spherical orbital labeled “S” and then two red and blue, peanut-shaped orbitals, each placed on **its own separate X, Y, Z axis coordinate system**, labeled “P subscript x” and “P subscript y.” The two red and blue orbitals are located on the x and y axes, respectively. The second diagram **shows three hybrid orbitals separately, each on its own X, Y, Z axis coordinate system. They are yellow and have one enlarged lobe and one smaller lobe, and are each oriented in a different direction in the X Y plane. The** third diagram shows the same three orbitals **together on the same X, Y, Z axis coordinate system. Their** larger lobes are located at and labeled as “120 degrees” from one another.

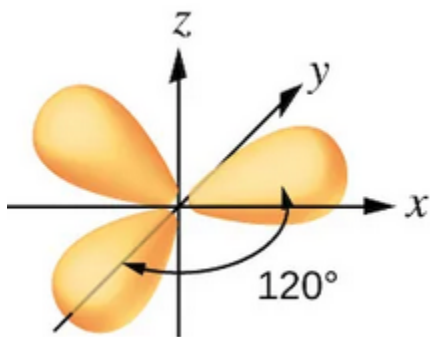
Chapter 8.2, section sp^2 Hybridization, 1st paragraph after figure 8.10

The paragraph reads:

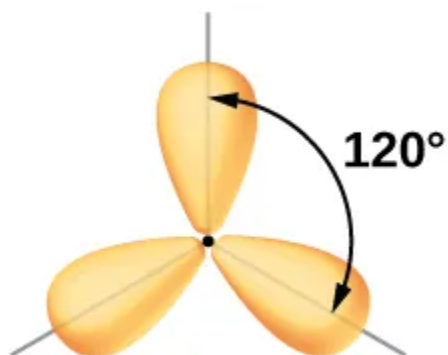
Although quantum mechanics yields the “plump” orbital lobes as depicted in Figure 8.10, sometimes for clarity these orbitals are drawn thinner and without the minor lobes, as in Figure 8.11, to avoid obscuring other features of a given illustration. We will use these “thinner” representations whenever the true view is too crowded to easily visualize.

I wonder if this text is old, because both figures 8.10 and 8.11 seem to be using the thinner orbital drawing style. For comparison:

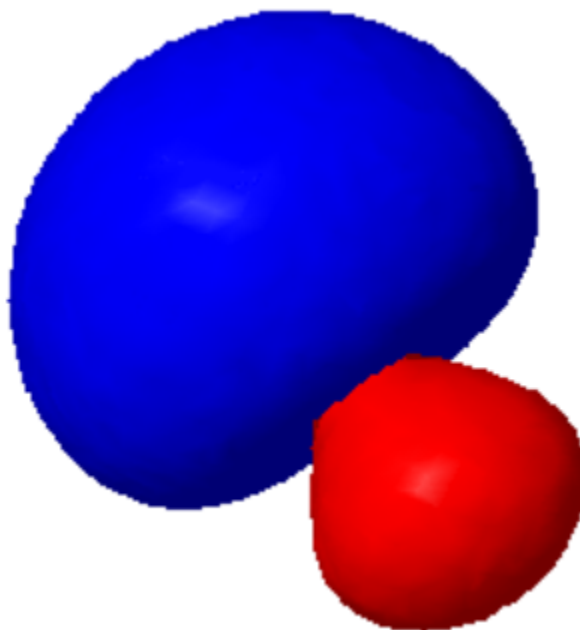
- Here is part of figure 8.10:



- Here is figure 8.11:



- Here is a plumper shape from [a page](#) on the site linked in the Link to Learning box just above.



- Here is a plumper shape, albeit of a different, non-hybridized orbital, from figure 6.21 in the book.

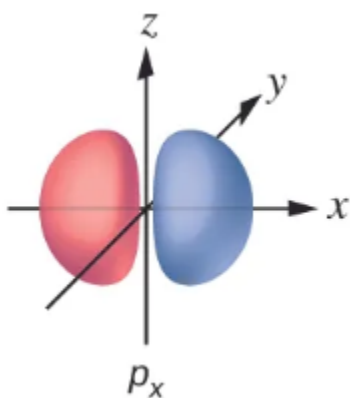


Figure 8.12 alt text

Sentence 1 reads (emphasis mine):

A boron atom is shown connected to three hydrogen atoms, which are arranged around it like a **pyramid**.

A pyramid is a 3 dimensional shape. There is no indication of three dimensionality in the figure. Also, the structure is explicitly “trigonal planar” from the caption, so I don’t know if “pyramid” is the best word to describe it. I suggest replacing the word “pyramid” with “**triangle**.”

Figure 8.13 alt text

The alt text does not mention the word “Unhybridized” by the single 2p orbital line on the right diagram.

Figure 8.15 caption

The figure 8.15 caption starts:

The hybridization of an s orbital (blue) and three p orbitals (red) ...

The p orbitals are not entirely red. I believe this is the same issue as the figure 8.8 and 8.10 captions.

Figure 8.15 alt text

The alt text reads:

A series of three diagrams connected by a right-facing arrow that is labeled, “Hybridization,” and a downward-facing arrow labeled, “Gives a tetrahedral arrangement,” are shown. The first diagram shows a blue spherical orbital and three red, peanut-shaped orbitals, each placed on an x, y, z axis system. The three red orbitals are located on the x, y and z axes, respectively. The second diagram shows the same four orbitals, but they are now purple and have one enlarged lobe and one smaller lobe. Each lies in a different axis in the drawing. The third diagram shows the same four orbitals, but their smaller lobes now overlap to form a tetrahedral structure.

Issues:

- Sentence 2 about the first diagram calls the red and blue p orbitals just red.
- Sentence 2 does not include the s, p_x , p_y , and p_z labels in the first diagram.
- Sentence 4 about the second diagram calls them “the same four orbitals”, but they are not the same.
- Sentence 4 also calls the second diagram orbitals purple when they are yellow.
- Sentence 6 says that in the second diagram, the hybrid orbitals “Each lies in a different axis in the drawing.” That could give the impression that the three hybrid orbitals lie along the x, y, and z axes, which they do not.
- The final sentence says it is the smaller lobes that form a tetrahedral structure in the third diagram, when in fact it is the bigger lobes.

A suggested replacement is (changes in bold):

A series of three diagrams **are shown** connected by a right-facing arrow that is labeled, "Hybridization," and a downward-facing arrow labeled, "Gives a tetrahedral arrangement." The first diagram shows **four orbitals, each drawn separately on its own x, y, z axis system. A** blue spherical orbital **is labeled "s."** Three red and blue, peanut-shaped orbitals **are labeled "p subscript x," "p subscript y," and "p subscript z," which** are located on the x, y and z axes, respectively. The second diagram **shows four different orbitals also drawn separately. These four orbitals are yellow** and have one enlarged lobe and one smaller lobe. Each lies in a different **direction** in the drawing. The third diagram shows the same four **yellow orbitals all drawn together on the same x, y, z axis system. Their larger lobes point in different directions** to form a tetrahedral structure.

Figure 8.17 alt text

The alt text reads:

Two diagrams are shown and labeled "a" and "b." Diagram a shows two carbon atoms, each surrounded by their four sp^3 hybridized orbitals in a three dimensional arrangement. Each of the orbitals is shown overlapping with a spherical hydrogen atom. Diagram b shows the same general arrangement, but the hydrogen atoms are just represented by an, "H" and their spherical orbitals are not shown.

Issues:

- In sentence 3, the sp^3 orbitals in part (a) are described as overlapping with spherical hydrogen atoms; they do not.
- The fact that one sp^3 orbital from one carbon atom overlaps an sp^3 orbital of another carbon atom is never mentioned.
- Sentence 4 describes the hydrogen atoms as just being represented by an "H" not spherical orbitals like they actually are.

Here is possible replacement alt text (changes in bold):

Two diagrams are shown and labeled "a" and "b." Diagram a shows two carbon atoms **next to each other**, each surrounded by their four **yellow** sp^3 hybridized orbitals in a three dimensional arrangement. **Six spherical blue hydrogen orbitals are shown around the carbon atoms.** Diagram b shows the same general arrangement, **but with orbitals overlapping. One yellow orbital from each carbon points toward the other carbon, and those orbitals partially overlap between the carbons. The remaining three yellow orbitals on each carbon partially overlap with spherical blue hydrogen orbitals.**

Figure 8.18 alt text

The alt text reads:

Three Lewis structures are shown along with designations of molecular shape. The left image shows a sulfur atom singly bonded to four fluorine atoms. The sulfur atom has

one lone pair of electrons while each fluorine has three. Two fluorine atoms are drawn vertically up and down from the sulfur while the other two are shown going into and out of the page. The second structure shows one chlorine atom singly bonded to three fluorine atoms. The chlorine has two lone pairs of electrons while each fluorine has three. Two fluorine atoms are drawn vertically up and down from the sulfur while the other is shown horizontally. The right structure shows a chlorine atom singly bonded to four fluorine atoms. The chlorine atom has one lone pair of electrons and a superscript plus sign, while each fluorine has three lone pairs of electrons. Two fluorine atoms are drawn vertically up and down from the sulfur while the other two are shown going into and out of the page.

I believe this is another one where the figure must have changed but the alt text did not get updated. Issues include:

- I don't see any "designations of molecular shape."
- The described lone pairs on the fluorine atoms are not present in the figure.
- The plus sign on ClF_4^+ is not on the Cl atom but outside unmentioned brackets.

Suggested replacement:

Three Lewis structures are shown. The left image shows a sulfur atom singly bonded to four fluorine atoms. The sulfur atom has one lone pair of electrons. Two fluorine atoms are drawn vertically up and down from the sulfur while the other two are shown going into and out of the page. The second structure shows one chlorine atom singly bonded to three fluorine atoms. The chlorine has two lone pairs of electrons. Two fluorine atoms are drawn vertically up and down from the sulfur while the other is shown horizontally. The right structure is surrounded by brackets followed by a superscript plus sign. It shows a chlorine atom singly bonded to four fluorine atoms. The chlorine atom has one lone pair of electrons. Two fluorine atoms are drawn vertically up and down from the sulfur while the other two are shown going into and out of the page.

Figure 8.19 alt text

The alt text reads:

Two images are shown and labeled "a" and "b." Image a depicts a ball-and-stick model in a trigonal bipyramidal arrangement. Image b depicts the hybrid orbitals in the same arrangement and each is labeled, "s p superscript three d."

This alt text is correct. However, understanding the shape described requires remembering the "trigonal bipyramidal" shape from a previous chapter. Also, it seems odd that the figure 8.18 alt text carefully described the bond directions while figure 8.19, which is even more focused on the three dimensional shape, omits that information.

Maybe something like this?

Two images are shown and labeled "a" and "b." Image a depicts a ball-and-stick model in a trigonal bipyramidal arrangement, where a central orange ball is surrounded by five green balls on sticks. Two green balls are vertically above and below the central ball. One green ball is to the right of the central ball. Two green balls are to the left of the central ball, one farther away in the third dimension and one closer. Image b depicts five

yellow hybrid orbitals sticking out from a central point in the same arrangement as the green balls from image a. Each orbital is labeled, “s p superscript three d.”

The figure 8.20 alt text has a similar issue.

Figure 8.20 alt text

The alt text reads:

Two images are shown and labeled “a” and “b.” Image a depicts a ball-and-stick model in an octahedral arrangement. Image b depicts the hybrid orbitals in the same arrangement and each is labeled, “s p superscript three d superscript two.”

This alt text is correct. However, like the figure 8.19 alt text, understanding the shape described requires remembering the “octahedral” shape from a previous chapter.

Maybe something like this?

Two images are shown and labeled “a” and “b.” Image a depicts a ball-and-stick model in an octahedral arrangement, where a central yellow ball is surrounded by six green balls on sticks. Two green balls are vertically above and below the central ball. The remaining four green balls are in a square on an implied plane sticking into and out of the paper through the central atom. Those four green balls are left toward the viewer, right toward the viewer, left away from the viewer, and right away from the viewer. Image b depicts six yellow hybrid orbitals sticking out from a central point in the same octahedral arrangement as the green balls from image a. Each orbital is labeled, “s p superscript three d superscript two.”

Chapter 8.2, section Assignment of Hybrid Orbitals to Central Atoms, numbered point 2

Numbered point 2 reads (emphasis mine):

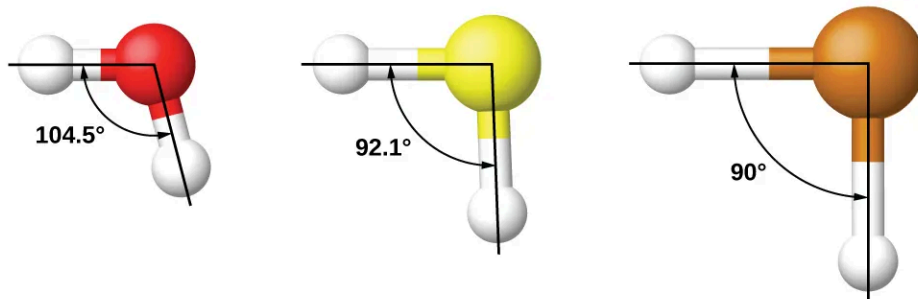
2. Determine the number of regions of electron density around an atom using VSEPR theory, in which single bonds, multiple bonds, **radicals**, and lone pairs each count as one region.

Would it be worth telling the reader what “radicals” are here? The term “radical” for single unpaired electrons does not appear until after this in Chapter 8.3. The term “free radical” was already introduced in Chapter 7.3 but that appears to refer to whole molecules, not single electrons.

What about (change in bold):

2. Determine the number of regions of electron density around an atom using VSEPR theory, in which single bonds, multiple bonds, **radicals (single unpaired electrons)**, and lone pairs each count as one region.

Unlabeled figure just above Example 8.2, alt text



The alt text reads:

Three Lewis structures are shown. The left structure shows an oxygen atom with two lone pairs of electrons single bonded to two hydrogen atoms. The middle structure is made up of a sulfur atom with two lone pairs of electrons single bonded to two hydrogen atoms. The right structure is made up of a tellurium atom with two lone pairs of electrons single bonded to two hydrogen atoms. From left to right, the bond angles of each molecule decrease.

Again, the diagram may have changed without the alt text being updated. Issues:

- Are these Lewis structures, or ball-and-stick models?
- The alt text describes lone pairs of electrons not present on the diagram.
- The alt text does not give the numeric angles included on diagram, which would quantify the “decrease” in angles and tie back in with the percentages given in the main just above.

Example 8.2, Solution section, 1st sentence wording

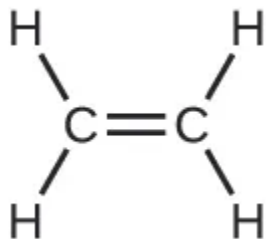
The first sentence starts:

The Lewis structure of sulfate shows ...

However, the unlabeled figure below that sentence appears to show a ball-and-stick model, not a Lewis structure?

Chapter 8.3

1st unlabelled figure in chapter 8.3, alt text



The second sentence reads (emphasis mine):

Each carbon atom is bonded to two hydrogen atoms by **a single bond**.
Since there are multiple hydrogen atoms bonded to each carbon atom, I think “a single bond” should be “**single bonds**”.

Figure 8.22

This may not be an error, but is there a reason why the 2s and 2p orbitals in this figure are only labeled as “s” and “p”? In both Figure 8.9 and in the Exercise 8.29 solution the 2s and 2p orbitals are labeled “2s” and “2p”, not just “s” and “p”. (I only noticed because I modeled my answer to Exercise 8.29 after Figure 8.22 so my answer was missing the “2s” and “2p” labels.)
Note: If the figure is changed, the alt text errata below may change slightly.

Figure 8.22 alt text

The alt text reads:

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left diagram shows an up-facing arrow labeled, “E.” To the lower right of the arrow is a short, horizontal line labeled, “2 s,” that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, “2 p.” Above both sets of lines is the phrase, “Orbitals in an isolated C atom.” Two of the lines have vertical, up-facing arrows drawn on them. The right side of the diagram shows three short, horizontal lines placed halfway up the space and each labeled, “s p superscript 2.” An upward-facing half arrow is drawn vertically on each line. Above these lines is one other short, horizontal line, labeled, “p.” Above both sets of lines is the phrase, “Orbitals in the s p superscript 2 hybridized C atom in C subscript 2 H subscript 4.”

Issues:

- The half arrow on the p orbital is missing from the description of the right diagram.
- The half arrows on the p orbitals in the left diagram are not described in the same stanza with the p orbitals themselves, being described instead after a break to discuss a label, making it harder to understand which of the orbital lines “Two of the lines” refers to.
- “Above both sets of lines is the phrase,” seems an odd way to describe the labels for the whole left and right sides of the diagram.
- The first part is described as “the left diagram.” The second part is described as “the right side of the diagram,” which is inconsistent with the first description.

Suggested replacement alt text (changes in bold):

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left **side of the** diagram shows an up-facing arrow labeled, “E.” To the lower right of the arrow is a short, horizontal line labeled, “2 s,” that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, “2 p,” **two of which have vertical, upward-facing arrows drawn on them. The left side of the diagram is labeled,** “Orbitals in an

isolated C **atom.**” **The** right side of the diagram shows three short, horizontal lines placed halfway up the space and each labeled, “s p superscript 2.” An upward-facing half arrow is drawn vertically on each line. Above these lines is one other short, horizontal line, labeled, “p.” **An upward-facing half arrow is drawn vertically on that line as well. The right side of the diagram is labeled,** “Orbitals in the s p superscript 2 hybridized C atom in C subscript 2 H subscript 4.”

Note: Exercise 8.31 solution unlabelled figure alt text has similar issues discussed in its own errata.

Figure 8.23 caption

Sentence 2 of the caption reads (emphases mine):

One C–C σ bond results from overlap of sp^2 hybrid **orbitals** on the carbon atom with one sp^2 hybrid **orbital** on the other carbon atom.

Taken literally, this says that multiple hybrid orbitals (plural) overlap with one hybrid orbital (singular).

Suggested replacement caption sentence 2 (changes in bold):

One C–C σ bond results from **the** overlap of **one** sp^2 hybrid **orbital** on **one** carbon atom with one sp^2 hybrid **orbital** on the other carbon atom.

Figure 8.23 alt text

The alt text reads:

Two diagrams are shown labeled, “a” and “b.” Diagram a shows two carbon atoms with three purple balloon-like orbitals arranged in a plane around them and two red balloon-like orbitals arranged vertically and perpendicularly to the plane. There is an overlap of two of the purple orbitals in between the two carbon atoms, and the other four purple orbitals that face the outside of the molecule are shown interacting with spherical blue orbitals from four hydrogen atoms. Diagram b depicts a similar image to diagram a, but the red, vertical orbitals are interacting above and below the plane of the molecule to form two areas labeled, “One pi bond.”

This alt text appears out of date. Issues include:

- There are no “purple” orbitals.
- The two p orbitals in (a) are not entirely red.
- The sigma bond and sigma bonds labels are not mentioned.
- The four hybridized orbitals that overlap with the spherical hydrogen orbitals are inconsistently described as “interacting”, not “overlapping.”
- This may not matter, but there is no mention that, in diagram “b”, the yellow orbitals in line between the carbon atoms are depicted as one yellow structure not two overlapping orbitals.

Here is possible replacement alt text:

Two diagrams are shown labeled, “a” and “b.” Diagram a shows two carbon atoms **on a geometric plane. Each carbon atom has** three **yellow** balloon-like orbitals arranged in

the plane and one dual-lobed peanut-shaped orbital arranged vertically, perpendicular to the plane. The dual-lobed orbitals each have a red lobe extending above the plane and a blue lobe extending below the plane. There is an overlap of two of the yellow orbitals in between the two carbon atoms. The other four yellow orbitals that face the outside of the molecule are shown overlapping with spherical blue orbitals from four hydrogen atoms. The overlaps are all labeled as sigma bonds. Diagram b depicts a similar image to diagram a, but the red lobes of the dual-lobed, vertical orbitals are connected with red above the plane, and the blue lobes are connected with blue below the plane. The red and blue areas above and below the plane of the molecule are labeled together as "One pi bond." In the plane, directly in line between the carbon atoms, the overlapping yellow orbitals are now shown as one large yellow blob in the plane.

Figure 8.24 alt text

The alt text reads:

A diagram of a carbon atom with two balloon-like purple orbitals labeled, "sp" arranged in a linear fashion around it is shown. Four red balloon-like orbitals are aligned in pairs in the y and z axes around the carbon and are labeled, "unhybridized p orbital," and, "Second unhybridized p orbital."

Again, this alt text appears old, mentioning purple orbitals. It also describes what I believe are two dual-lobed (red and blue lobed) p orbitals as four orbitals instead of two.

Here is suggested replacement alt text (changes in bold), although perhaps the wording could be clarified further:

A diagram of a carbon atom with two balloon-like yellow orbitals labeled, "sp" arranged in a linear fashion around it is shown. There are also two dual-lobed red and blue orbitals, each with a red lobe on one side of the carbon atom and a blue lobe on the other side. If a line through the yellow orbitals is considered the x axis, the dual-lobed orbitals are arranged along the y and z axes, perpendicular to the two yellow orbitals and each other. The dual-lobed orbitals are labeled, "unhybridized p orbital," and, "Second unhybridized p orbital."

Note: At this point, I was considering starting a conspiracy theory about how the Chemistry Illuminati are hiding the existence of the mysterious purple orbitals, but I decided against it.

Figure 8.25 alt text

The alt text reads:

Two diagrams are shown and labeled, "a" and "b." Diagram a shows two carbon atoms with two purple balloon-like orbitals arranged in a plane around each of them, and four red balloon-like orbitals arranged along the y and z axes perpendicular to the plane of the molecule. There is an overlap of two of the purple orbitals in between the two carbon atoms. The other two purple orbitals that face the outside of the molecule are shown interacting with spherical blue orbitals from two hydrogen atoms. Diagram b depicts a

similar image to diagram a, but the red, vertical orbitals are interacting above and below and to the front and back of the plane of the molecule to form two areas labeled, "One pi bond," and, "Second pi bond," each respectively.

Oh no, more purple orbitals. Suggested replacement (changes in bold):

Two diagrams are shown and labeled, "a" and "b." Diagram a shows two carbon atoms with two **yellow** balloon-like orbitals arranged in a **line** around each of them, and four red **and blue dual-lobed** orbitals arranged along the y and z axes perpendicular to the **line** of the molecule. There is an overlap of two of the **yellow** orbitals in between the two carbon atoms. The other two **yellow** orbitals that face the outside of the molecule are shown interacting with spherical blue orbitals from two hydrogen atoms. Diagram b depicts a similar image to diagram a, but the red **and blue dual-lobed** vertical orbitals are **connected** above and below and to the front and back of the **line** of the molecule. **One pair of red and blue regions across from each other is** labeled, "One pi bond," and **the other pair is labeled**, "Second pi bond."

Chapter 8.4

Figure 8.27 alt text

Sentence 3 reads (emphasis mine):

On the left there is a line attached to a test tube labeled, "**Sample tube.**"

This may not matter, but the test tube is actually just labeled "**Sample,**" without the word "tube" in the label.

Figure 8.28 alt text

The alt text reads:

A pair of diagrams are shown and labeled, "a" and "b." Diagram a shows two identical waves with two crests and two troughs. They are drawn one above the other with a plus sign in between and an equal sign to the right. To the right of the equal sign is a much taller wave with a same number of troughs and crests. Diagram b shows two waves with two crests and two troughs, but they are mirror images of one another rotated over a horizontal axis. They are drawn one above the other with a plus sign in between and an equal sign to the right. To the right of the equal sign is a flat line.

The alt text is correct except for a minor grammatical issue where "a same" should be "the same". However, I think an additional phrase would make it even clearer. I suggest the following (changes in bold):

A pair of diagrams are shown and labeled, "a" and "b." Diagram a shows two identical waves with two crests and two troughs. They are drawn one above the other with a plus sign in between and an equal sign to the right. To the right of the equal sign is a much taller wave with **the** same number of troughs and crests. Diagram b shows two waves with two crests and two troughs, but they are mirror images of one another rotated over

a horizontal axis, **such that the crests of one wave line up with the troughs of the other wave**. They are drawn one above the other with a plus sign in between and an equal sign to the right. To the right of the equal sign is a flat line.

Figure 8.29 caption

The first sentence of the caption reads:

Sigma (σ) and sigma-star (σ^*) molecular orbitals are formed by the combination of two s atomic orbitals.

Why are these called “Sigma (σ) and sigma-star (σ^*)” here without an “s”? Shouldn’t they be called “Sigma-s (σ_s) and sigma-s-star (σ_s^*)”, with “s” included? The current first sentence as a whole might be inaccurate as written, because p orbitals can also combine to form sigma molecular orbitals according to Figure 8.30.

Figure 8.29 alt text

The alt text reads (phrases to be removed in bold):

A diagram is shown that depicts a vertical upward-facing arrow that lies to the left of all the other portions of the diagram and is labeled, “E.” To the immediate right of the midpoint of the arrow are two circles each labeled with **a positive sign**, the letter S, and the phrase, “Atomic orbitals.” These are followed by a right-facing horizontal arrow that points to the same two circles **labeled with plus signs**, but they are now touching and are labeled, “Combine atomic orbitals.” Two right-facing arrows lead to the last portion of the diagram, one facing upward and one facing downward. The upper arrow is labeled, “Subtract,” and points to two oblong ovals labeled with **plus signs**, and the phrase, “Antibonding orbitals sigma subscript s superscript asterisk.” The lower arrow is labeled, “Add,” and points to an elongated oval **with two plus signs** that is labeled, “Bonding orbital sigma subscript s.” The heading over the last section of the diagram are the words, “Molecular orbitals.”

Issues:

- The alt text mentions plus signs that are not in the diagram.
- The alt text does not include the nuclei dots or, more importantly, how the orbital shapes in the rightmost column are positioned relative to the nuclei.
- The alt text says the right top figure is labeled “Antibonding orbitals ...” plural. The diagram label in fact starts “Antibonding orbital ...” singular.

Suggested replacement alt text (changes other than deletions in bold):

A diagram is shown that depicts a vertical upward-facing arrow that lies to the left of all the other portions of the diagram and is labeled, “E.” To the immediate right of the midpoint of the arrow are two circles, **each with a dot in the center, and each labeled with the letter S. The phrase, “Atomic orbitals” appears above the two circles.** These are followed by a right-facing horizontal arrow that points to the same two circles, but they are now touching and are labeled, “Combine atomic orbitals.” Two right-facing arrows lead to the last portion of the diagram, one facing upward and one facing

downward. The upper arrow is labeled, “Subtract,” and points to two oblong ovals labeled with the phrase, “Antibonding **orbital** sigma subscript s superscript asterisk.” **The oblong ovals are not centered on the nuclear dots, but are shifted away from each other avoiding the space between the dots.** The lower arrow is labeled, “Add,” and points to an elongated oval that is labeled, “Bonding orbital sigma subscript s.” **This elongated oval includes both nuclear dots and the space between them.** The heading over the last section of the diagram are the words, “Molecular orbitals.”

Chapter 8.4, 2nd Link to Learning, wrong link

The 2nd Link to Learning appears just after Figure 8.29. It includes a link on the word “animations” which is supposed to take you to an Orbitron website, but instead takes you to a YouTube video about sigma and pi atomic (not molecular) orbitals. Searching for the Orbitron website finds this link: <https://winter.group.shef.ac.uk/orbitron/>. However, that page states that it is incomplete and does not include molecular orbitals. It sounds like it was being redone without Flash, but is incomplete.

Figure 8.30 alt text

The alt text reads:

Two horizontal rows of diagrams are shown. The upper diagram shows two equally-sized peanut-shaped orbitals with a plus sign in between them connected to a merged orbital diagram by a right facing arrow. The merged diagram has a much larger oval at the center and much smaller ovular orbitals on the edge. It is labeled, “sigma subscript p x.” The lower diagram shows two equally-sized peanut-shaped orbitals with a plus sign in between them connected to a split orbital diagram by a right facing arrow. The split diagram has a much larger oval at the outer ends and much smaller ovular orbitals on the inner edges. It is labeled, “sigma subscript p x superscript asterisk”.

The phase colors of the orbitals are not mentioned. I think that information is crucial to understand the different resulting shapes due to constructive and destructive interference. Also, giving the location of the nuclei represented as dots may be important to contextualize where the electron density is relative to the contributing atoms’ nuclei.

Here is a possible replacement:

Two horizontal rows of diagrams are shown. Both rows contain a two-lobed orbital with a lobe on each side of a nuclear dot, a plus sign, another two-lobed orbital centered on another nuclear dot, a right facing arrow, and a result orbital diagram. The rows differ in the phase colors of the lobes of the input orbitals and the shapes of the result orbitals. More specifically, the top row shows that an orbital with a blue left lobe and a red right lobe plus an orbital with a red left lobe and a blue right lobe results in a three-lobed orbital. The resulting orbital has a big red oval in the center between the two nuclear dots and small blue ovals on the outsides, to the left of the left nuclear dot and to the right of the right nuclear dot. The three-lobed result orbital is labeled “sigma subscript p x.” The bottom row shows that an orbital with a red left lobe and a blue right lobe plus an

orbital with a red left lobe and a blue right lobe results in a four-lobed orbital. The resulting orbital has a big red lobe to the left of the left nuclear dot, a small blue lobe to the right of the left nuclear dot, an area between the nuclear dots with no lobes, a small red lobe to the left of the right nuclear dot, and a big blue lobe to the right of the right nuclear dot. The four-lobed result is labeled “sigma subscript p x superscript asterisk.”

Figure 8.31 alt text

The alt text reads:

Two horizontal rows of diagrams are shown. The upper and lower diagrams both begin with two vertical peanut-shaped orbitals with a plus sign in between followed by a right-facing arrow. The upper diagram shows the same vertical peanut orbitals bending slightly away from one another and separated by a dotted line. It is labeled, “pi subscript p superscript asterisk.” The lower diagram shows the horizontal overlap of the two orbitals and is labeled, “pi subscript p.”

This does not address the phase colors, which are crucial because they explain why the top and bottom rows are different instead of identical. Not all nodal planes are described. The last sentence might also make the location of the overlap clearer.

Suggested replacement alt text:

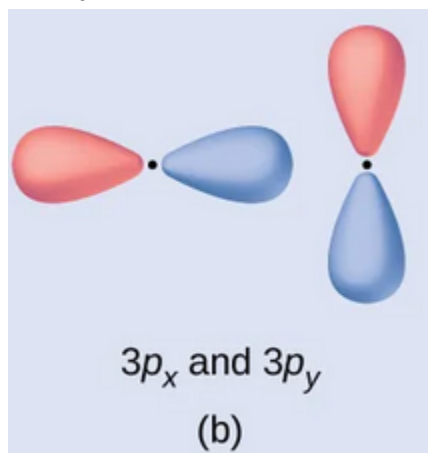
Two horizontal rows of diagrams are shown. The upper and lower diagrams both begin with two vertical peanut-shaped orbitals with a plus sign in between followed by a right-facing arrow. The top row orbitals have opposite colors, adding an orbital with the top lobe red and the bottom lobe blue to an orbital with the top lobe blue and the bottom lobe red. After the arrow, the upper diagram shows the same vertical peanut-shaped orbitals bending slightly away from one another. It is labeled, “pi subscript p superscript asterisk.” Two nodal planes are shown. One plane is through the nuclei and perpendicular to the lobes, extending left and right as well as into and out of the paper. The other plane separates the nuclei and the lobes, extending up and down as well as into and out of the paper. The lower row starts with two vertical peanut-shaped orbitals with identical colors, adding an orbital with the top lobe red and the bottom lobe blue to another orbital with the top lobe red and the bottom lobe blue. After the arrow, the bottom diagram shows the upward-facing red lobes of the two peanut-shaped vertical orbitals are connected with red, and the downward-facing blue lobes are connected with blue. Only one nodal plane is shown. It runs through both nuclei, perpendicular to the red and blue regions above and below and below the plane. The plane extends left and right as well as into and out of the paper. There is no vertical plane separating the nuclei.

Example 8.5, Solution section (b)

Answer (b) reads (emphasis mine):

(b) will not result in a new orbital **because the in-phase component (bottom) and out-of-phase component (top) cancel out**. Only orbitals with the correct alignment can combine.

Is the text in bold accurate? I see them not aligning, but I do not see opposite phases aligning to cancel. The two orbital shapes from the question part (b) are included below for reference. Looking at that, I see no regions where red on one shape overlaps with blue on the other shape, or vice versa. The fact that the contributing orbitals are not aligned (as the answer even says) would seem to preclude cancellation, or at least perfect cancellation. I wonder if the text in bold could just be removed.



Chapter 8.3, Portrait of a Chemist box

Ok, I hate to point this out, but the first sentence says Walter Kohn “is”, while the last sentence says that Walter Kohn passed away. Should the “is” in the first sentence technically be “was”?

Figure 8.34 figure or alt text

Unlike Figure 8.35, Figure 8.34 does not have the headings “Atomic orbitals”, “Molecular orbitals”, and “Atomic orbitals” along the top of the diagram. The alt text for Figure 8.34 includes them, but the figure itself does not. I don’t know if the figure should be fixed or just the alt text.

Possible resolutions:

- Add the headings to Figure 8.34 and remove the ungrammatical word “is” from the last sentence of the Figure 8.34 alt text, or
- Leave Figure 8.34 as is and remove the entire last sentence describing the headers from the Figure 8.34 alt text.

Figure 8.37 alt text

The alt text reads:

A graph is shown in which the y-axis is labeled, “E,” and appears as a vertical, upward-facing arrow. Across the top, the graph reads, “L i subscript 2,” “B e subscript 2,” “B subscript 2,” “C subscript 2,” “N subscript 2,” “O subscript 2,” “F subscript 2,” and “Ne subscript 2.” Directly below each of these element terms is a single pink line, and all

lines are connected to one another by a dashed line, to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled, “sigma subscript 2 p x superscript asterisk”. Directly below each of these lines is a set of two pink lines, and all lines are connected to one another by a dashed line, to create an overall line that decreases in height as it moves from left to right across the graph. It is consistently lower than the first line. This line is labeled, “pi subscript 2 p y superscript asterisk,” and, “pi subscript 2 p z superscript asterisk.” Directly below each of these double lines is a single pink line, and all lines are connected to one another by a dashed line, to create an overall line that decreases in height as it moves from left to right across the graph. It has a distinctive drop at the label, “O subscript 2.” This line is labeled, “sigma subscript 2 p x.” Directly below each of these lines is a set of two pink lines, and all lines are connected to one another by a dashed line to create an overall line that decreases very slightly in height as it moves from left to right across the graph. It is consistently lower than the third line until it reaches the point labeled, “O subscript 2.” This line is labeled, “pi subscript 2 p y,” and, “pi subscript 2 p z.” Directly below each of these lines is a single blue line, and all lines are connected to one another by a dashed line to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled, “sigma subscript 2 s superscript asterisk.” Finally, directly below each of these lines is a single blue line, and all lines are connected to one another by a dashed line to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled. “sigma subscript 2 s.”

Issues:

- The lines are black, not pink and blue.
- Sentences 8 and 11 begin the descriptions of the two lines that cross over at oxygen. They both start with “Directly below ...”. However, in those two sentences, that is only true for part of the diagram. Therefore, in the suggested replacement alt text below, I suggest replacing the “directly below ...” phrases with the word “next” only in those two sentences. Then, I clarify the relationship when discussing how it crosses over at oxygen.
- There are a couple of other places where I think an additional word or two would make things clearer.

Suggested replacement alt text (changes in bold):

A graph is shown in which the y-axis is labeled, “E,” and appears as a vertical, upward-facing arrow. Across the top, the graph reads, “L i subscript 2,” “B e subscript 2,” “B subscript 2,” “C subscript 2,” “N subscript 2,” “O subscript 2,” “F subscript 2,” and “Ne subscript 2.” Directly below each of these element terms is a single **black horizontal** line, and all lines are connected to one another by a dashed line, to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled, “sigma subscript 2 p x superscript asterisk”. Directly below each of these lines is a set of two **black lines side by side**, and all lines are connected to one another by a dashed line, to create an overall line that decreases in height as it moves from left to right across the graph. It is consistently lower than the first line. This line is labeled, “pi subscript 2 p y superscript asterisk,” and, “pi subscript 2 p z superscript asterisk.” **Next** is a single **black** line, and all lines are connected to one another by a dashed line, to

create an overall line that decreases in height as it moves from left to right across the graph. It has a distinctive drop at the label, " σ_2 ." This line is labeled, " σ_x ." **Next** is a set of two **black** lines **side by side**, and all lines are connected to one another by a dashed line to create an overall line that decreases very slightly in height as it moves from left to right across the graph. **This line is labeled, " π_y ," and, " π_z ."** It is consistently lower than the **previously mentioned " σ_x " line** until it reaches the **column** labeled, " σ_2 ," **at which point the " σ_x " line drops below the " π_y ," and, " π_z " line.** **Directly below is a single black line**, and all lines are connected to one another by a dashed line to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled, " σ_{s^*} ." Finally, directly below each of these lines is a single **black** line, and all lines are connected to one another by a dashed line to create an overall line that decreases in height as it moves from left to right across the graph. This line is labeled. " σ_s ."

Figure 8.38 alt text

Issues:

- The upward facing y-axis arrow labeled E is not described in the alt text.
- The alt text describes vertical arrows that are not present on the diagram.

I'm not even going to try to suggest a replacement for this monster.

Figure 8.40 alt text

Sentence 5 contains the phrase "two vertical **half-arrow**" (singular). It should be "two vertical **half-arrows**" (plural).

The alt text reads:

A diagram is shown that has an upward-facing vertical arrow running along the left side labeled, "E." At the bottom center of the diagram is a horizontal line labeled, " σ_s ," that has two vertical half arrows drawn on it, one facing up and one facing down. This line is connected to the right and left by upward-facing, dotted lines to two more horizontal lines, each labeled, " 2_s ," and with two vertical half arrows drawn on them, one facing up and one facing down. These two lines are connected by upward-facing dotted lines to another line in the center of the diagram, but farther up from the first and labeled, " σ_{s^*} ." This horizontal line has two vertical half-arrow drawn on it, one facing up and one facing down. Moving further up the center of the diagram is a horizontal line labeled, " σ_x ," which lies below two horizontal lines, lying side-by-side, and labeled " π_y ," and " π_z ." Both the bottom and top lines are connected to the right and left by upward-facing, dotted lines to three more horizontal lines, each labeled, " 2_p ," on either side. These sets of lines each hold three

upward-facing and one downward-facing half-arrow. They are connected by upward-facing dotted lines to another single line and then pair of double lines in the center of the diagram, but farther up from the lower lines. They are labeled, “sigma subscript 2 p subscript x superscript asterisk,” “pi subscript 2 p subscript y superscript asterisk,” and “pi subscript 2 p subscript z superscript asterisk,” respectively. The lower of these two central, horizontal lines each contain one upward-facing half-arrow. The left and right sides of the diagram have headers that read, “Atomic orbitals,” while the center header reads, “Molecular orbitals.”

That is an impressively good description that nevertheless has a few issues:

- The vertical half arrows on the σ_{2px} , π_{2py} , and π_{2pz} lines are not included.
- Everything is described bottom to top until the σ_{2px}^* , π_{2py}^* , and π_{2pz}^* lines at the top. There, the top σ_{2px}^* line is described first, and its vertical relationship to the π_{2py}^* , and π_{2pz}^* lines is not really described. I suggest describing these lines bottom to top as well and making the vertical relationship explicit.
- Sentence 5 contains the phrase “two vertical **half-arrow**” (singular). It should be “two vertical **half-arrows**” (plural).
- Sentence 9 contains the phrase “then pair of double lines.” It should be “then **a** pair of double lines” with the added word “a”.
- I don’t understand why the phrases “half-arrow” have hyphens but the phrases “half arrows” (plural) do not have hyphens. Then again, maybe that is correct. Since I didn’t know, I kept the existing convention in my suggested replacement alt text below.

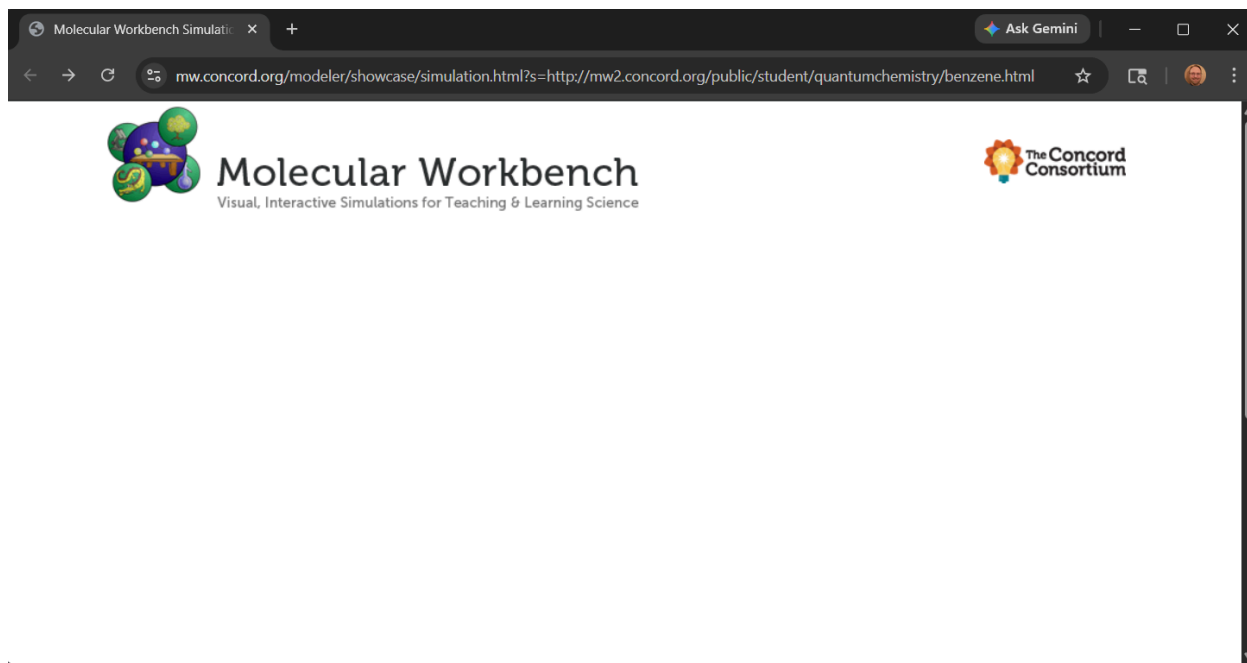
Suggested replacement alt text (changes in bold):

A diagram is shown that has an upward-facing vertical arrow running along the left side labeled, “E.” At the bottom center of the diagram is a horizontal line labeled, “sigma subscript 2 s,” that has two vertical half arrows drawn on it, one facing up and one facing down. This line is connected to the right and left by upward-facing, dotted lines to two more horizontal lines, each labeled, “2 s,” and with two vertical half arrows drawn on them, one facing up and one facing down. These two lines are connected by upward-facing dotted lines to another line in the center of the diagram, but farther up from the first and labeled, “sigma subscript 2 s superscript asterisk.” This horizontal line has two vertical **half arrows** drawn on it, one facing up and one facing down. Moving further up the center of the diagram is a horizontal line labeled, “sigma subscript 2 p subscript x,” **with two vertical half arrows drawn on it, one facing up and one facing down. That horizontal line** lies below two horizontal lines, lying side-by-side, and labeled “pi subscript 2 p subscript y,” and “pi subscript 2 p subscript z,” **with two vertical half arrows drawn on each, one facing up and one facing down.** Both the bottom and top lines are connected to the right and left by upward-facing, dotted lines to three more horizontal lines, each labeled, “2 p,” on either side. These sets of lines each hold three upward-facing and one downward-facing half-arrow. They are connected by upward-facing dotted lines to **a pair of two horizontal lines and an even higher single line** in the center of the diagram. **These two central, horizontal lines are labeled, “pi subscript 2 p subscript y superscript asterisk,” and each contains one upward-facing half-arrow. The highest single central line is labeled “sigma subscript 2 p subscript x superscript asterisk,” and has no vertical arrows.** The left and right

sides of the diagram have headers that read, "Atomic orbitals," while the center header reads, "Molecular orbitals."

Chapter 8.4, 3rd Link to Learning box, bad link

The 3rd Link to Learning box, at the very bottom of Chapter 8.4, has a "three-dimensional drawings" link. The link works, but the resulting page is empty with only a logo but no contents. I tested on Firefox and Chrome on Windows 11.



Chapter 8 - Key Terms

radical (suggested new term)

What about adding **radical** as a key term for Chapter 8. One reason I mention it is that **free radical** is a key term for Chapter 7, so it might not be bad to disambiguate the two terms by having radical as a key term in its own right, especially since one seems to refer to an unpaired electron and the other to a molecule with unpaired electrons (if I'm understanding correctly). The term radical appears at least twice in Chapter 8:

- Chapter 8.2, section Assignment of Hybrid Orbitals to Central Atoms, numbered point 2
- Chapter 8.3, body text just after Figure 8.25

linear combination of atomic orbitals

According to Chapter 8.2, in the body text paragraph below Figure 8.6, sentence 4, LCAO is not only used to calculate molecular orbitals, but also hybridized atomic orbitals. Perhaps hybridized atomic orbitals are just a subset of molecular orbitals and the existing definition is still correct. However, I mention it just in case the definition of LCAO should include a mention of hybrid orbitals in addition to molecular orbitals.

Chapter 8 - Exercises

Note: Only odd numbered exercises were fully reviewed as I do not have instructor access.

Exercise 8.3 solution

The book solution reads:

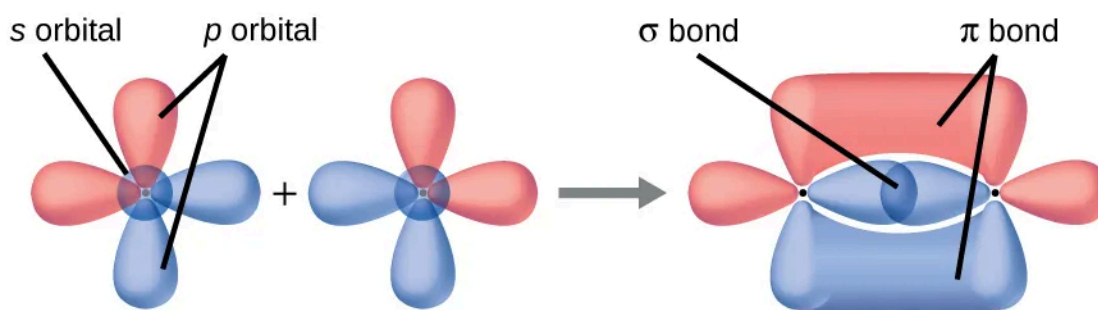
The specific average bond distance is the distance with the lowest energy. At distances less than the bond distance, the positive charges on the two nuclei repel each other, and the overall energy increases.

Should the solution also mention the repulsion between all of the two atoms' electrons?

Chapter 8.1, the paragraph just before Figure 8.2, second to last sentence expresses it this way (emphasis mine):

If the distance between the nuclei were to decrease further, the repulsions between nuclei **and the repulsions as electrons are confined in closer proximity** to each other would become stronger than the attractive forces.

Exercise 8.5 solution, unlabeled figure



Why are the s orbitals shown on the left but not the right. It's not like they disappear, they just aren't involved in the bonding, correct? Shouldn't they be present or absent in both? If a change is made to the figure, the alt text should be changed, too.

Exercise 8.5 solution, unlabeled figure alt text

The alt text reads:

This figure shows the orbitals of two atoms being added together as they form bonds. The two atoms are shown separately on the right, each having two peanut-shaped orbitals lying perpendicularly to one another. A right-facing arrow shows that the two have moved closer together and now the upper and lower portions of the vertical peanut-shaped orbitals are shown as merging together above and below the plane of the molecule while the horizontal peanut-shaped orbitals are overlapping between the two nuclei.

Issues:

- Sentence two says the two atoms are shown separately on the **right**. It should be **left**.
- Is “above and below the plane of the molecule” a good description when there is no plane shown? The “plane” of the molecule could be interpreted as the plane of the paper in this case, instead of a horizontal plane going into and out of the paper.
- I would describe the vertical peanut shape orbitals more as “connected” than “overlapping”, since the original shapes of the orbitals on the left would still not even touch while on the right. However, maybe that is a limitation of the figure more than an issue with the truth of the alt text.
- The labels are missing.

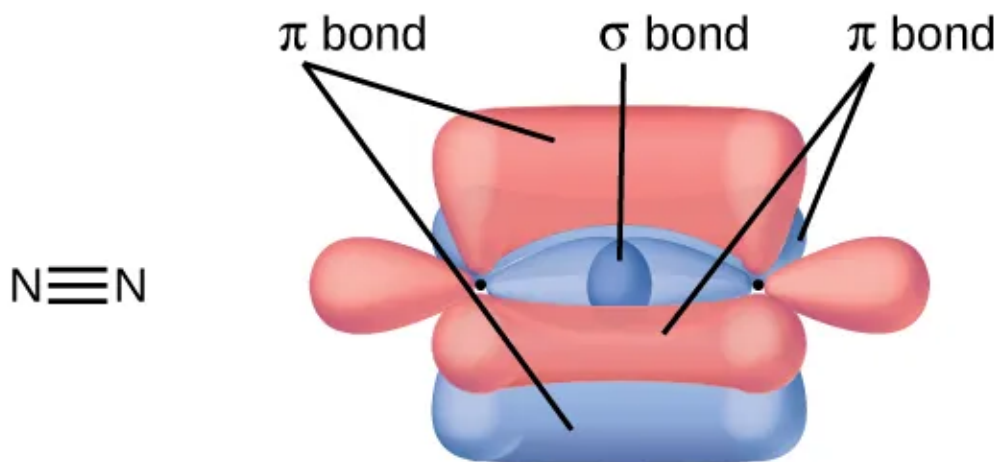
Here is a suggested replacement. See also preceding issue on s orbitals being present on the left in the figure itself but not on the right:

This figure shows the orbitals of two atoms being added together as they form bonds. The two atoms are shown separately on the **left, separated by a plus sign. Each atom has one small spherical orbital labeled “s orbital,”** and two peanut-shaped orbitals lying perpendicularly to one another. **The label “p orbital,” has lines pointing to two lobes of one peanut-shaped orbital. After the two atoms is a right-facing arrow and an image of the two atoms with bonds. The atoms have moved closer together and now the ends of the horizontal peanut-shaped orbitals overlap end-to-end between the atoms. This overlap is labeled “sigma bond.”** The upper and lower lobes of the vertical peanut-shaped orbitals are shown **as connected both** above and below the **sigma bond. The label “pi bond” has lines pointing to both the upper and lower connections joining the vertical orbitals.**

Exercise 8.7 solution

Why does the solution include a figure when the question does not ask the student to draw anything? It just asks a yes or no question of whether you agree. Also, should the question ask if you agree and also why? The book solution includes information on the number of different bond types, but the student might not give those in an answer because it is just a yes or no question.

Exercise 8.7 solution, unlabeled image alt text



Sentences 4 and 5 of the alt text read:

One of the orbitals lies horizontally and overlaps between the two nuclei. It is labeled, "sigma bond."

Read literally, this says that one orbital overlaps itself, which reminds me of one hand clapping. How about replacing alt text sentences 4 and 5 with this (changes in bold):

The horizontal orbitals from each atom overlap between the two nuclei. **This overlap** is labeled, "sigma bond."

Exercise 8.13 question

The solution includes the counts of lone pairs for each case, but the question never asks for that information. Should the question also ask for counts of lone pairs?

Exercise 8.13 solution

The book answer reads:

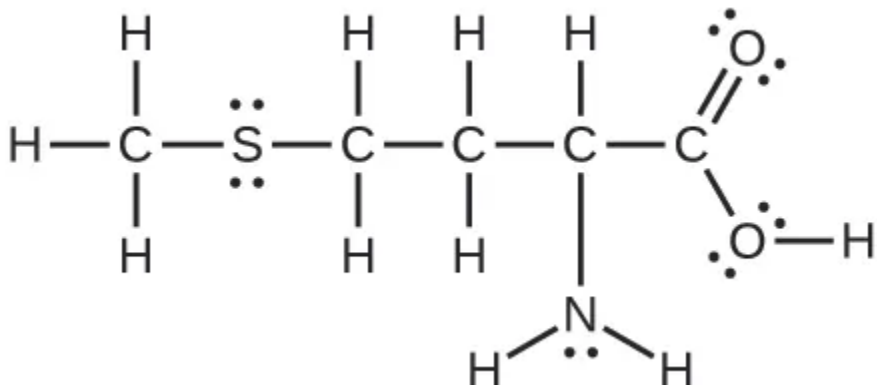
trigonal planar, sp^2 ; trigonal pyramidal (one lone pair on A) sp^3 ; T-shaped (two lone pairs on A) sp^3d , or (three lone pairs on A) sp^3d^2

The second open parenthesis is missing its closing parenthesis. Also, the question is in 4 parts, while the answer has 3 semicolon-delimited top level divisions, and the last one covers two cases because the shape part of the answer, "T-shaped" is the same for both. This is easy to spot if you already know the last two parts give the same shape. However, if a student gets that wrong, and is trying to understand the correct answer, the way it appears to have three answers to four questions could be confusing.

I suggest correcting the parenthesis and just having four top-level portions to the answer and repeating the shape that applies to the two of them (changed portion in bold):

trigonal planar, sp^2 ; trigonal pyramidal (one lone pair on A) sp^3 ; T-shaped (two lone pairs on **A**) **sp^3d** ; **T-shaped** (three lone pairs on A) sp^3d^2

Exercise 8.14 question, unlabeled image alt text



Sentence 4 of the alt text starts:

The fourth **andfinal** carbon ...

The “andfinal” should be two words “**and final**”.

Exercise 8.15 (b) solution, unlabeled figure alt text



The alt text reads (emphasis mine):

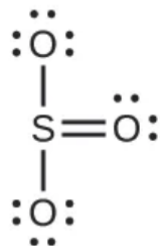
Two Lewis **structure** are shown, connected by a double-ended arrow. The left structure shows a sulfur atom with one lone pair of electrons double bonded to an oxygen atom with two lone pairs of electrons on the left and single bonded to an oxygen atom with three lone pairs of electrons on the right. The right structure shows the same molecule, except that the double bonded oxygen is on the right side of the sulfur and the single bonded oxygen is to the left of the sulfur.

In the first sentence, the word “structure” should be “**structures**.” Also, the Exercise 7.45 (a) solution has the same image. I don’t know if this is essential, but the 7.45 (a) solution’s alt text includes an additional sentence noting that the lone pairs shift between left and right along with the oxygen bonds. (Note that the 7.45 (a) solution’s alt text is not perfect in another way; separate errata submitted). Here is that additional sentence:

The lone pairs of electrons have also shifted to account for the change of bond types.

Exercise 8.15 (b) and (c) solutions, unlabeled figures

(c) Trigonal planar, sp^2



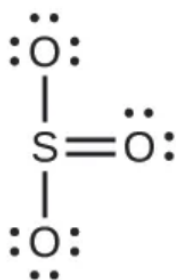
(b) Bent (120°), sp^2



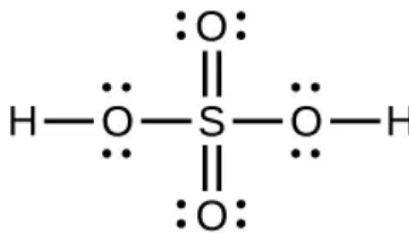
This may not matter, but the 8.15 (b) solution image includes both resonance forms of SO_2 , while the 8.15 (c) solution image does not include the resonance forms of SO_3 . (This assumes I am correct about SO_3 having two additional resonance forms, of course.)

Exercise 8.15 (c) and (d) solutions, unlabeled figures

(c) Trigonal planar, sp^2



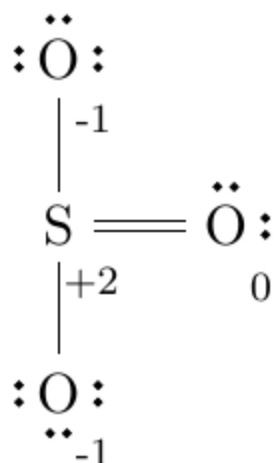
(d) Tetrahedral, sp^3



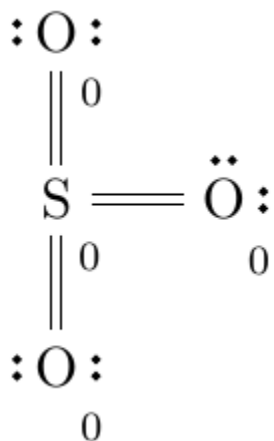
I may be missing something, but I do not think the book has covered why the sulfur atom in the Exercise 8.15 (c) SO_3 molecule is not hypervalent but the sulfur atom in the Exercise 8.15 (d) H_2SO_4 molecule is hypervalent. I am not suggesting the book answers are wrong. I am suggesting that the book text might be missing something that explains the difference. To illustrate the difference I am seeing, I will walk through my logic following the same procedure for both 8.15 (c) and 8.15 (d), and where I found the explanations in the book.

Exercise 8.15 (c), SO_3 :

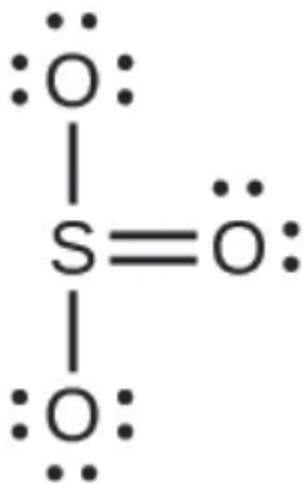
First, I follow the steps in Chapter 7.3, section Writing Lewis Structures with the Octet Rule, observing the octet rule. Then I label the formal charges based on Chapter 7.4, section Calculating Formal Charge to get this (with help from the LaTeX chemfig package for the graphics)...



Second, I follow Chapter 7.3, section Exceptions to the Octet Rule, subsection Hypervalent Molecules to observe that sulfur is a period three atom, and therefore could have more than 8 valence electrons. Making it hypervalent allows me to further reduce formal charges like this...

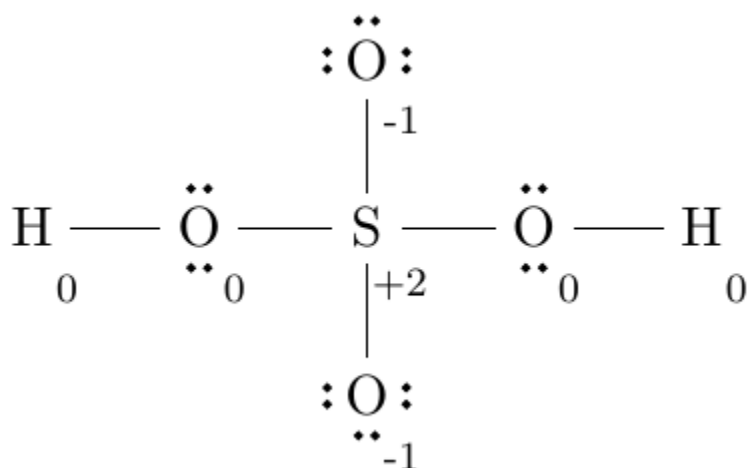


In the case of Exercise 8.15 (c), the first Lewis diagram, which observes the octet rule, is correct, and the second Lewis diagram, which is hypervalent, is wrong...

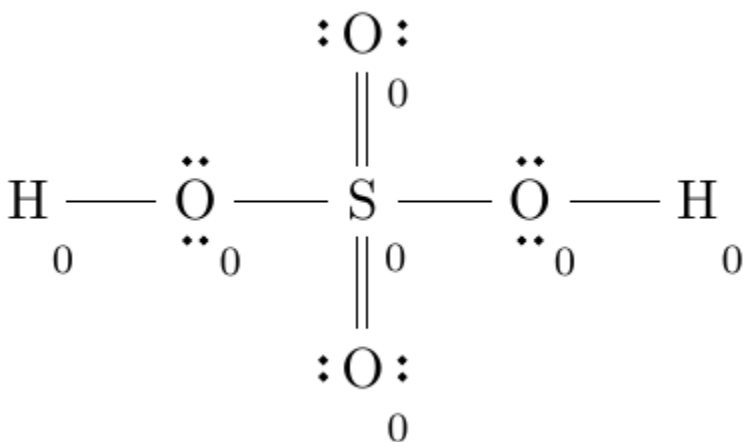


Exercise 8.15 (d), H₂SO₄:

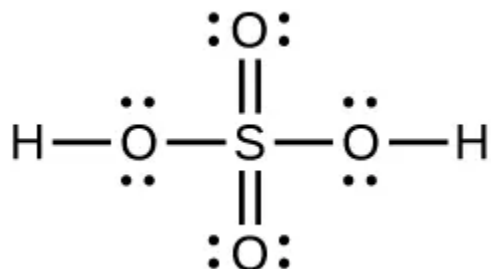
First, I follow the same first step as for Exercise 8.15 (c), observing the octet rule, which gives me this...



Second, I follow the same second step as for Exercise 8.15 (c), making it hypervalent to reduce formal charges, which gives me this...



In the case of Exercise 8.15 (d), the second Lewis diagram, which is hypervalent, is correct, and the first Lewis diagram, which observes the octet rule, is wrong.



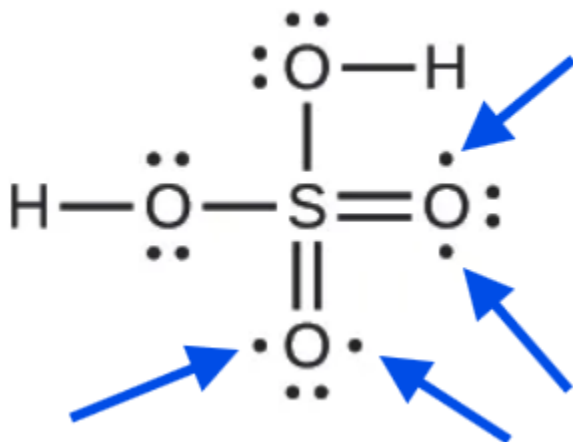
In other words, the same procedure that works for Exercise 8.15 (d) does not work for Exercise 8.15 (c), and I cannot find anywhere in the first 8 chapters of the book that explains why these are different. When does the octet rule take precedence, and when does reducing formal

charges take precedence? As I mentioned, I may be missing something. However, I am submitting this errata in case the book is not explaining some step required to do the exercise.

Note: This is basically the same question I had as in my Chapter 7 document errata on Exercise 7.63 solution, unpaired electrons, bullet point three.

Trivial Note: Just to confuse things further, the Exercise 7.63 solution Lewis structure for the H_2SO_4 molecule differs from the Exercise 8.15 (d) solution Lewis structure for the same molecule in that it has unpaired electrons (see blue arrows below). However, the Exercise 7.63 solution has the same bonding structure as the Exercise 8.15 (d) solution, so I don't think the difference is relevant to this errata. I created a separate errata for the unpaired electrons in my Chapter 7 errata document.

63.



Exercise 8.15 (d) solution, unlabeled figure alt text

The alt text reads:

A Lewis structure is shown in which a sulfur atom is single bonded to four oxygen atoms.

Two of the oxygen atoms have three lone pairs of electrons while the other two each have two lone pairs of electrons and are each singly bonded to a hydrogen atom.

The first sentence is wrong in saying that all the bonds to the oxygen atoms are single. The second sentence is wrong in saying that any oxygen atoms have three lone pairs of electrons. Here is suggested replacement alt text:

A Lewis structure is shown in which a sulfur atom is **double** bonded to **two** oxygen atoms **and single bonded to two more oxygen atoms**. The **double bonded** oxygen atoms **each** have **two** lone pairs of electrons. The **single bonded oxygen atoms** each have two lone pairs of electrons and are each singly bonded to a hydrogen atom.

Exercise 8.19 question text

Three issues:

- Question part (b) asks you to describe the geometry. However, it does not specify whether it is asking for the electron pair geometry (including all regions of electron density) or the molecular shape geometry (including only bonds, not free lone pairs).
- Both questions part (b) and part (c) include the phrase “the P atoms, the **S atom**”. Since there are multiple sulfur atoms, shouldn’t they say “**S atoms**” plural?
- Question part (d) includes the phrase “the oxidation states and formal **charge**”. Why is “oxidation states” plural and “formal charge” singular? Shouldn’t it be “formal **charges**” plural?

Exercise 8.19 question unlabeled image alt text



The alt text reads:

A Lewis structure is shown in which three phosphorus atoms are single bonded together to form a triangle. Each phosphorus is bonded to a sulfur atom by a vertical single bond and each of those sulfur atoms is then bonded to a single phosphorus atom so that a six-sided ring is created with a sulfur in the middle.

This is all correct but I think it could be clearer. In particular, “each of those sulfur atoms is then bonded to a single phosphorus atom” could be interpreted to mean that each sulfur atom is bonded to a different phosphorus atom instead of all the sulfur atoms being bonded to the same phosphorus atom. Also, it might not hurt to tell the reader there are no lone pairs shown.

I will offer two possible replacement alt texts, one replacement that fixes those issues with minimal changes, and another replacement that is completely reworded to put the ring shape in the reader’s mind first instead of last.

Minimal replacement alt text (changes in bold);

A Lewis structure is shown **without its lone pairs** in which three phosphorus atoms are single bonded together to form a triangle. Each phosphorus is bonded to a sulfur atom by a vertical single bond and **all** of those sulfur atoms **are** then bonded to a single **fourth** phosphorus atom so that a six-sided ring is created with a sulfur in the middle.

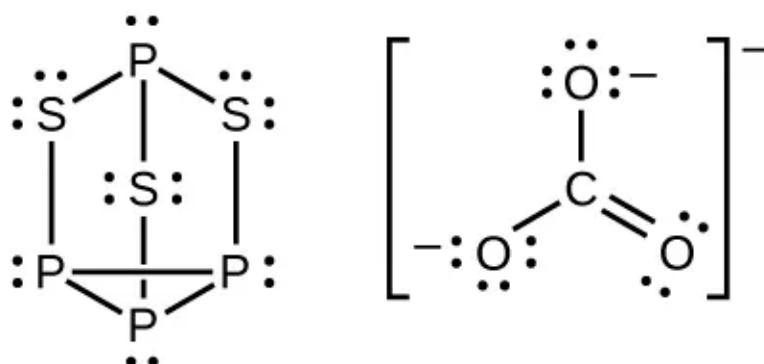
Completely reworded replacement alt text:

A Lewis structure is shown without its lone pairs. It has a six-sided ring with a phosphorus atom at the top, sulfur atoms at the top left and top right, phosphorus atoms

at the bottom left and bottom right, and a phosphorus atom at the bottom. There is an additional sulfur atom in the middle of the ring which is bonded to both the top phosphorus and the bottom phosphorus. The bottom left phosphorus is also bonded directly to the bottom right phosphorus across the width of the ring. In this two dimensional diagram, that bond appears to cross the bond between the bottom phosphorus and sulfur atom in the middle of the ring.

See also the separate errata on the alt text for the Exercise 8.19 solution. Part of that image is the same, so the alt text updates should probably be coordinated.

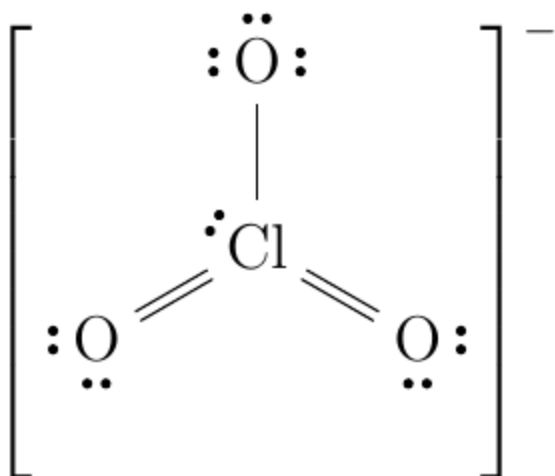
Exercise 8.19 (a) solution unlabeled image



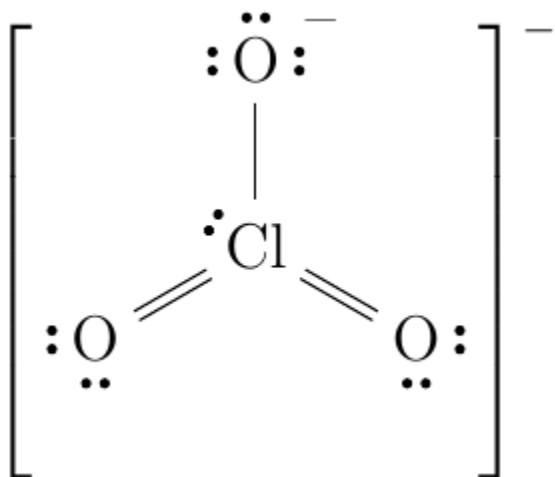
There are multiple issues with the right part of the figure, the Lewis structure for ClO_3^- :

- The right image has a C for carbon in the center. Since the ion is ClO_3^- , it should have a Cl for chlorine in the center instead.
- The total number of valence electrons is wrong. 7 for Cl, plus 3 times 6 for O, plus 1 for the ion charge, which gives 26. The book solution has 2 single bonds for 4, plus one double bond for 4, plus 8 lone pairs on O for 16, which gives 24.
- If the negative signs by two of the oxygen atoms are formal charges, then the total formal charges on the atoms, -2, do not add up to the overall formal charge on the ion, which is -1.
- If the negative signs by the oxygen atoms are formal charges, note that the question did not explicitly ask for formal charges. Furthermore, I believe the text and examples in the book only cover putting formal charges on all atoms, not just some, and using labels like -1, 0, +1, not just negative signs. Those are found in Chapter 7.4 section Using Formal Charge to Predict Molecular Structure. See also my Chapter 7 errata document, section Lewis structures with only some atoms labeled positive or negative.

Shouldn't the right side of the diagram look more like this?



Or, if you want to include the negative sign by the oxygen atom with the non-zero formal charge, then wouldn't it look like this?



Note: The replacement alt text in the alt text errata below does not include all of these changes.

Exercise 8.19 (a) solution unlabeled image alt text

The alt text reads:

Two Lewis structure are shown, the left of which depicts three phosphorus atoms single bonded together to form a triangle. Each phosphorus is bonded to a sulfur atom by a vertical single bond and each of those sulfur atoms is then bonded to a single phosphorus atom so that a six-sided ring is created with a sulfur in the middle. Each sulfur atom in this structure has two lone pairs of electrons while each phosphorus has one lone pair. The second Lewis structure shows a chlorate atom single bonded to two oxygen atoms, each of which has three lone pairs of electrons, and double bonded to an oxygen atom, which has two lone pairs of electrons.

Issues:

- Sentence 1 starts “Two Lewis structure are shown ...” The word “structure” should be “structures”.
- The fact that all the bonds in the first structure are single bonds is never stated.
- The phrase “each of those sulfur atoms is then bonded to a single phosphorus atom” could mean that each sulfur atom is bonded to a different phosphorus atom instead of them all being bonded to the same phosphorus atom.
- Sentence 4 says there is a “chlorate atom” in the second structure when it should say “chlorine atom”. (Image shows a C for carbon not a Cl for chlorine, but I believe that is in error and there is another errata including that error.)
- The description of the second structure does not include the minus signs by two of the oxygens, the brackets, or the minus sign outside the brackets.

Part of this alt text reproduces the alt text for the image in the Exercise 8.19 question. In that errata, I offered two replacement alt texts for consideration, one minimally fixing the issues, and the other rewording the description of the first structure to describe the ring shape first. Here are two possible replacement alt texts that are analogous to those in the errata for the Exercise 8.19 question unlabeled image alt text.

Note: These replacements do not take into account all the changes suggested to the figure itself in the figure errata above.

Minimal replacement alt text (changes in bold);

Two Lewis **structures** are shown, the left of which depicts three phosphorus atoms single bonded together to form a triangle. Each phosphorus is bonded to a sulfur atom by a vertical single bond and **all** of those sulfur atoms **are** then bonded to a single **fourth** phosphorus atom so that a six-sided ring is created with a sulfur in the middle. **All bonds in this structure are single bonds.** Each sulfur atom in this structure has two lone pairs of electrons while each phosphorus has one lone pair. The second Lewis structure shows a **chlorine** atom single bonded to two oxygen atoms, each of which has three lone pairs of electrons **and a minus sign by it. The chlorine atom is also** double bonded to **another** oxygen atom, which has two lone pairs of electrons. **The entire second structure is surrounded by a pair of brackets. There is a superscripted minus sign after the right bracket.**

Replacement alt text with the description of the first structure completely reworded:

Two Lewis **structures** are shown, the left of which depicts **a six-sided ring with a phosphorus atom at the top, sulfur atoms at the top left and top right, phosphorus atoms at the bottom left and bottom right, and a phosphorus atom at the bottom. There is an additional sulfur atom in the middle of the ring which is bonded to both the top phosphorus and the bottom phosphorus. The bottom left phosphorus is also bonded directly to the bottom right phosphorus across the width of the ring. In this two dimensional diagram, that bond appears to cross the bond between the bottom phosphorus and sulfur atom in the middle of the ring. All bonds in this structure are single bonds.** Each sulfur atom in this structure has two lone pairs of electrons while each phosphorus has one lone pair. The second Lewis structure shows a **chlorine** atom single bonded to two oxygen atoms, each of which has three lone pairs of electrons **and a minus sign by it. The chlorine atom is also** double

bonded to **another** oxygen atom, which has two lone pairs of electrons. **The entire second structure is surrounded by a pair of brackets. There is a superscripted minus sign after the right bracket.**

See also the separate errata on the alt text for the Exercise 8.19 question. Part of this image is the same as that image, so the alt text updates should probably be coordinated.

Exercise 8.19 solution

Issues:

- Solution part (a) - See separate errata above on Exercise 8.19 (a) solution unlabeled image.
- Solution part (b), S atoms portion - For the S atoms, the solution includes “two lone pairs” in the answer. Lone pair counts were not asked for. I understand that the lone pair count influences how much less than 109.5 degrees the bent angle will be, but I don’t know if students have been shown an example before where they need to include a free pair count as part of a geometry question.
- Solution part (b), Cl atom portion - This portion says “Cl atoms”, but since there is only one, shouldn’t it be “Cl atom”? More importantly, this answer should probably match the figure in part (a). See errata above on Exercise 8.19 (a) solution unlabeled image. The book solution for part (b) gives “trigonal pyramidal” as the geometry for the ClO_3^- ion, which may be correct if that other errata updates the Lewis diagram in part (a). However, the existing Lewis diagram in part (a) seems to show a trigonal planar geometry instead because it has 3 bonds but no lone pairs on the central atom. That Lewis structure has issues though.
- Solution part (d) - I do not believe students have been given the knowledge to answer the oxidation state part of the question for all of these atoms. More specifically, the rules in the Chapter 4.2 section on Oxidation-Reduction Reactions covers hydrogen, oxygen, and halogens. The rules do not cover phosphorus or sulfur directly, and I don’t think there is enough information from other atoms to use rule 4 which says “The sum of oxidation numbers for all atoms in a molecule or polyatomic ion equals the charge on the molecule or ion.” Also, nothing in those rules covers oxidation states in increments of $\frac{1}{2}$ such as the $-1\frac{1}{2}$ answer given for sulfur.

Exercise 8.21 solution unlabeled image alt text



Sentence 3 of the alt text starts (emphasis mine):

The right structure shows a phosphorus **atoms** ...

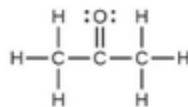
The word “atoms” should be “**atom**”.

Exercise 8.30 question formatting

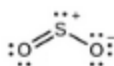
The unlabelled Lewis structure figures in the question are very small and hard to read. This is on the web version, tested with both Chrome and Firefox browsers on Windows 11. (Note: I think they may appear bigger in this screenshot than they do on my screen, and they are still small.)

30. For each of the following structures, determine which electrons will be delocalized:

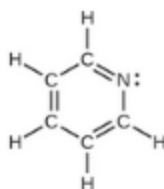
(a) Hybridization of each carbon



(b) Hybridization of sulfur



(c) All atoms



Exercise 8.30 (b) question unlabeled image alt text

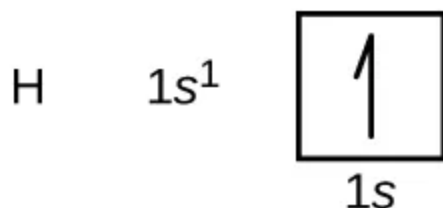
The alt text begins (emphasis mine):

A Lewis structure is shown in which a sulfur atom with **two lone pairs** of electrons and a positive sign ...

The “two lone pairs” of electrons in that opening phrase should be “**one lone pair**”.

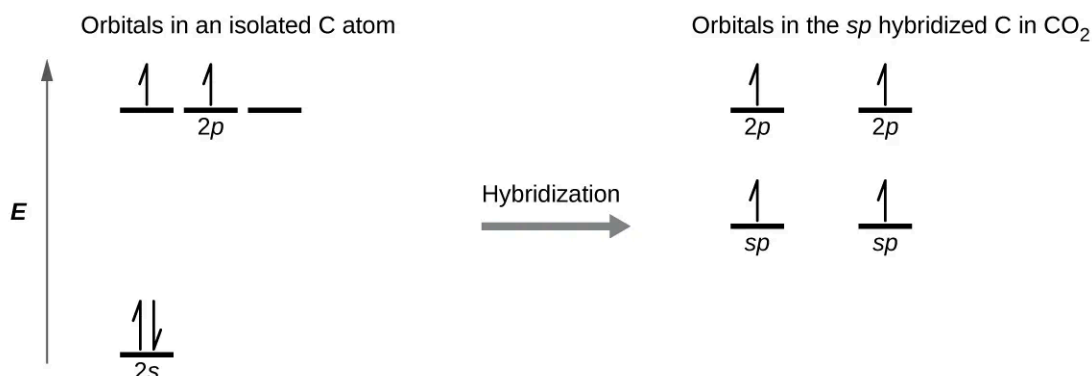
Exercise 8.31 question

The question asks for an “orbital diagram”, which the Chapter 6.4 section The Aufbau Principle, 2nd paragraph defines as a diagram that looks like this:



In Chapter 8.2, section *sp* Hybridization, 2nd paragraph and the Figure 8.9 caption, diagrams that look like the Exercise 8.31 solution are named as “orbital energy-level diagram”, not just “orbital diagram”. Therefore, shouldn’t “**orbital diagram**” in the question text be replaced with “**orbital energy-level diagram**”?

Exercise 8.31 solution unlabeled figure alt text



The alt text reads:

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left diagram shows an up-facing arrow labeled, “E.” To the lower right of the arrow is a short, horizontal line labeled, “2 s,” that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, “2 p.” Above both sets of lines is the phrase, “Orbitals in an isolated C atom.” There are two upward facing arrows on two of these lines. The right side of the diagram shows two short, horizontal lines placed halfway up the space and each labeled, “s p.” An upward-facing half arrow is drawn vertically on each line. Above these lines are two other short, horizontal lines, each labeled, “2 p,” and which have two upward facing arrows on them. Above both sets of lines is the phrase, “Orbitals in the s p hybridized C in C O subscript 2.”

Issues:

- The half arrows on the p orbitals in the left diagram are not described in the same stanza with the p orbitals themselves, being described instead after a break to discuss a label, making it harder to understand which of the orbital lines “Two of the lines” refers to.
- “Above both sets of lines is the phrase,” seems an odd way to describe the labels for the whole left and right sides of the diagram.
- The second-to-last sentence discusses two lines, ‘each labeled, “2 p,” and which have two upward facing arrows on them.’ This could be misinterpreted as meaning each line has two upward facing arrows on it.
- The first part is described as “the left diagram.” The second part is described as “the right side of the diagram,” which is slightly inconsistent in wording..

Suggested replacement alt text (changes in bold):

A diagram is shown in two parts, connected by a right facing arrow labeled, “Hybridization.” The left **side of the** diagram shows an up-facing arrow labeled, “E.” To

the lower right of the arrow is a short, horizontal line labeled, "2 s," that has two vertical half-arrows facing up and down on it. To the upper right of the arrow are a series of three short, horizontal lines labeled, "2 p," **two of which have vertical, upward-facing arrows drawn on them. The left side of the diagram is labeled,** "Orbitals in an isolated C atom." There are two upward facing arrows on two of these lines. The right side of the diagram shows two short, horizontal lines placed halfway up the space and each labeled, "s p." An upward-facing half arrow is drawn vertically on each line. Above these lines are two other short, horizontal lines, each labeled, "2 p," and **each of which has an** upward facing arrows on it. **The right side of the diagram is labeled,** "Orbitals in the s p hybridized C in C O subscript 2."

Note: figure 8.22 alt text has similar issues discussed in its own errata.

Exercise 8.33 (b) solution

The second part of the book solution for part (b), the part describing how ψ is different between atomic and molecular orbitals, reads:

Differences: ψ for an atomic orbital describes the behavior of only one electron at a time based on the atom. For a molecule, ψ represents a mathematical combination of atomic orbitals.

Is the first sentence correct? Doesn't a wave function describe the same number of electrons in either case? Put another way, doesn't the count of molecular orbitals (bonding and antibonding) equal the count of atomic orbitals that were combined to make it, making the number of electrons held the same?

Exercise 8.41 solution

The solution is correct, but it contains non-integer bond orders. The bond order formula given in Chapter 8.4 section Bond Order can produce non-integer results, so it makes sense. However, I don't think the chapter includes any examples that introduce the reader to the possibility of non-integer bond orders. It might be nice if there were an example in the text that gave a non-integer bond order as a result.